

# **Erie County (Buffalo) Sample — Shopping Center / Retail**

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*Traffic Impact Study*

Prepared August 18, 2026 · 42.9780, -78.7900

**SIS sample draft — full tier**

# Traffic Impact Study

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## TRAFFIC IMPACT STUDY FOR DOT PROJECTS

Prepared in Accordance With Chapter 5 of the NYSDOT Highway Design Manual (HDM)

|                     |                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| Project name        | Erie County (Buffalo) Sample — Shopping Center / Retail                                    |
| PIN                 | PIN xxxx.xx (to be assigned by NYSDOT)                                                     |
| Project description | Shopping Center / Retail (85 1,000 sqft GLA) within Buffalo-Cheektowaga-Niagara Falls MSA. |
| NYSDOT Region       | Region 5 — Buffalo / Western New York                                                      |
| Site coordinates    | 42.9780°, -78.7900°                                                                        |
| Opening year (ETC)  | 2027                                                                                       |

### Note:

It is a violation of law for any person, unless they are acting under the direction of a licensed professional engineer, architect, landscape architect, or land surveyor, to alter an item in any way. If an item bearing the stamp of a licensed professional is altered, the altering engineer, architect, landscape architect, or land surveyor shall stamp the document and include the notation "altered by" followed by their signature, the date of such alteration, and a specific description of the alteration.

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

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**NYSDOT region:** Region 5 — Buffalo / Western New York

**Planning group:** GBNRTC (Greater Buffalo-Niagara Regional Transportation Council)

**Regulatory regime:** SEQRA (ECL Article 8 / 6 NYCRR Part 617).

**SEQRA classification:** Not yet determined — lead agency to classify as Type I (6 NYCRR Part 617.4 — likely EIS), Type II (Part 617.5 — exempt), or Unlisted (Part 617.6 — Short EAF + significance finding) before locking the TIS.

## 1.0 SUMMARY OF TRAFFIC IMPACTS

### Capacity:

A capacity analysis was performed per NYSDOT Highway Design Manual Chapter 5. The following capacity improvement measures are cost effective, and are recommended to reduce reoccurring delay: Minor: signal-timing optimization (shift 3-5s of green to the critical phase) is sufficient. No geometric change required; Minor: signal-timing optimization (shift 3-5s of green to the critical phase) is sufficient. No geometric change required; Major: add a dedicated turn lane on the critical approach AND retune the signal; reconsider site driveway alignment or development scale if delay remains above 80s; Major: add a dedicated turn lane on the critical approach AND retune the signal; reconsider site driveway alignment or development scale if delay remains above 80s; Major: add a dedicated turn lane on the critical approach AND retune the signal; reconsider site driveway alignment or development scale if delay remains above 80s; Major: add a dedicated turn lane on the critical approach AND retune the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

### Crash:

A minimum three-year accident history review was performed and the safety screening met all of the required steps. A full crash analysis is not required.

### GML §239-m / §239-n county planning board referral required:

The site is within 500 ft of a state or county classified roadway, which triggers mandatory referral to the controlling county planning board under General Municipal Law §239-m (site-plan / special-permit / variance / zoning amendment) or §239-n (subdivision review) before the local board may act. The county may make adverse findings under §239-f; the local board can override only by

supermajority. Practical TIS consequence: the county will typically request the TIS in full as part of the \$239 referral package — pre-application coordination with the county planning agency is recommended.

|                           |                       |                        |                              |
|---------------------------|-----------------------|------------------------|------------------------------|
| <b>9</b><br>INTERSECTIONS | <b>0</b><br>LOS DROPS | <b>4</b><br>AT LOS E/F | <b>6.2s</b><br>WORST Δ DELAY |
|---------------------------|-----------------------|------------------------|------------------------------|

## 2.0 TRAVEL SPEEDS

Per NYSDOT HDM Chapter 5 §5.2, posted speed limits and actual operating speeds (85th-percentile) along the study network are reported below. Posted-speed-limit values in the third column are auto-ingested from the NYSDOT Roadway Data Management (RDM) FeatureServer ([https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM\\_Roadway\\_Current/FeatureServer](https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM_Roadway_Current/FeatureServer)) at PDF-generation time — the same source data viewable through the NYSDOT Traffic Data Viewer (<https://gis.dot.ny.gov/html5viewer/?viewer=tdv>). Operating speeds (85th-percentile) are NOT published by NYSDOT and require a project-specific radar / floating-car speed study.

| Street                            | Functional class                    | Posted speed | 85th-pctile operating speed |
|-----------------------------------|-------------------------------------|--------------|-----------------------------|
| Getzville Road (GETZVILLE RD)     | Urban Major Collector               | 30 mph       | Speed study required        |
| Near Sheridan Drive               | Urban Principal Arterial Other      | 40 mph       | Speed study required        |
| Harlem Road                       | Urban Principal Arterial Interstate | 55 mph       | Speed study required        |
| Near Harlem Road (HARLEM RD)      | Urban Minor Arterial                | 35 mph       | Speed study required        |
| Sheridan Drive                    | Urban Principal Arterial Other      | 40 mph       | Speed study required        |
| Millersport Highway (HARTFORD RD) | Urban Local                         | 30 mph       | Speed study required        |

## 3.0 CAPACITY ANALYSIS

Capacity analyses performed in this report are consistent with HCM 6th Edition. The software used to perform this analysis is the SimpleImpactStudies HCM 6 solver (per-approach control-delay model). Synchro / SimTraffic, HCS, Sidra, or VISSIM may be substituted for formal submittal per NYSDOT Regional Traffic Office preference.

Study network functional-class composition (per NYSDOT RDM): Urban Principal Arterial Other (4); Urban Major Collector (2); Urban Principal Arterial Interstate (1); Urban Minor Arterial (1); Urban Local (1). NYSDOT functional class drives access management expectations under HDM Appendix 5A (entrance standards for state highways) and the default K-factor banding under HDM §5.3.

Level of Service thresholds applied below are per HCM 2010 Exhibits 18-4 (signalized), 19-1 / 20-2 / 21-1 (unsignalized — 2-way stop / all-way stop / roundabout), and 10-7 (freeway). Any component  $v/c > 1.0$  is LOS F regardless of delay or density.

**Signalized intersections (HCM 2010 Exhibit 18-4, p. 18-6) — control delay sec/veh; any  $v/c > 1.0$  is F.**

| LOS     | A   | B     | C     | D     | E     | F   |
|---------|-----|-------|-------|-------|-------|-----|
| sec/veh | ≤10 | 10-20 | 20-35 | 35-55 | 55-80 | >80 |

**Unsignalized — 2-way stop / all-way stop / roundabout (HCM 2010 Exhibits 19-1, 20-2, 21-1) — sec/veh; any  $v/c > 1.0$  is F.**

| LOS     | A   | B     | C     | D     | E     | F   |
|---------|-----|-------|-------|-------|-------|-----|
| sec/veh | ≤10 | 10-15 | 15-25 | 25-35 | 35-50 | >50 |

**Freeway basic / weaving / merge-diverge (HCM 2010 Exhibit 10-7, p. 10-9) — density pc/mi/ln; any component  $vd/c > 1.00$  is F.**

| LOS | A | B | C | D | E | F |
|-----|---|---|---|---|---|---|
|-----|---|---|---|---|---|---|

| LOS      | A   | B     | C     | D     | E     | F   |
|----------|-----|-------|-------|-------|-------|-----|
| pc/mi/ln | ≤11 | 11-18 | 18-26 | 26-35 | 35-45 | >45 |

### 3.1 Growth Rates

Background traffic growth is applied at 1.5% per year over 1 year. This rate is the per-station median CAGR measured from the NYSDOT Traffic Monitoring AADT FeatureServer for Region 5 — Buffalo / Western New York: 0.13%/yr median across 3,171 count stations (rolling actual-count window approximately 2000-2019; 25th-75th percentile range -1.24% to 1.48%). The measured value supersedes the NYSDOT statewide default fallbacks (per HDM Chapter 5: 1.7%/yr Interstate + ramps from the Data Services Bureau, 2.0%/yr all other roads from the Regional Planning Group). The controlling Regional Planning Group GBNRTC (Greater Buffalo-Niagara Regional Transportation Council) may have separately-adopted growth-rate guidance that overrides the measured median for formal submittal — confirm at scoping.

### 3.2 Existing AADT and DHV

Existing AADT is sourced from the NYSDOT Traffic Data Viewer. Design Hourly Volume (DHV) is reported per HDM Chapter 5 convention as AADT × K-factor; this analysis applies K = 0.10 as the statewide design default — facility-specific K should be substituted from NYSDOT count-station records for formal submittal. The table below grows the existing volume at 1.5%/yr to Estimated Time of Completion (ETC), ETC + 20 years, and ETC + 30 years per HDM Chapter 5 §5.3.

| Intersection                          | 2026<br>AADT | 2026<br>DHV | 2027<br>(ETC)<br>AADT | 2027<br>(ETC)<br>DHV | 2047<br>AADT | 2047<br>DHV | 2057<br>AADT | 2057<br>DHV |
|---------------------------------------|--------------|-------------|-----------------------|----------------------|--------------|-------------|--------------|-------------|
| Getzville Road & Sheridan Drive       | 12,421       | 1,242       | 12,606                | 1,261                | 16,980       | 1,698       | 19,706       | 1,971       |
| Near Sheridan Drive                   | 12,420       | 1,242       | 12,606                | 1,261                | 16,979       | 1,698       | 19,705       | 1,970       |
| Harlem Road & Sheridan Drive          | 12,421       | 1,242       | 12,607                | 1,261                | 16,980       | 1,698       | 19,706       | 1,971       |
| Near Harlem Road                      | 5,279        | 528         | 5,358                 | 536                  | 7,217        | 722         | 8,375        | 838         |
| Sheridan Drive & Youngmann Expressway | 2,361        | 236         | 2,396                 | 240                  | 3,228        | 323         | 3,746        | 375         |
| Millersport Highway & Sheridan Drive  | 21,820       | 2,182       | 22,147                | 2,215                | 29,829       | 2,983       | 34,618       | 3,462       |
| Millersport Highway & Sheridan Drive  | 21,819       | 2,182       | 22,146                | 2,215                | 29,828       | 2,983       | 34,616       | 3,462       |
| Sheridan Drive & Frankhauser Road     | 33,011       | 3,301       | 33,506                | 3,351                | 45,128       | 4,513       | 52,373       | 5,237       |
| Millersport Highway & Fiint Road      | 24,039       | 2,404       | 24,400                | 2,440                | 32,863       | 3,286       | 38,139       | 3,814       |

Source: AADT estimated from engine peak-hour entering-volume sum via K = 0.10. NYSDOT Traffic Data Viewer measured AADT is recommended for submittal. Projection applies the 1.5%/yr background growth rate compounded annually.

### 3.3 Traffic Control Device Data

Existing traffic control devices at each study-network intersection are summarized below. Signal timing data (cycle length, phase splits, offset) and per-approach detection presence should be confirmed against the controlling NYSDOT Regional Traffic Office or municipal traffic-engineering signal record for formal submittal.

| Intersection                          | Control type                        | Approaches w/ detection     |
|---------------------------------------|-------------------------------------|-----------------------------|
| Getzville Road & Sheridan Drive       | Signal (controlling RTO to confirm) | Field verification required |
| Near Sheridan Drive                   | Signal (controlling RTO to confirm) | Field verification required |
| Harlem Road & Sheridan Drive          | Signal (controlling RTO to confirm) | Field verification required |
| Near Harlem Road                      | Signal (controlling RTO to confirm) | Field verification required |
| Sheridan Drive & Youngmann Expressway | Signal (controlling RTO to confirm) | Field verification required |
| Millersport Highway & Sheridan Drive  | Signal (controlling RTO to confirm) | Field verification required |
| Millersport Highway & Sheridan Drive  | Signal (controlling RTO to confirm) | Field verification required |

| Intersection                      | Control type                        | Approaches w/ detection     |
|-----------------------------------|-------------------------------------|-----------------------------|
| Sheridan Drive & Frankhauser Road | Signal (controlling RTO to confirm) | Field verification required |
| Millersport Highway & Flint Road  | Signal (controlling RTO to confirm) | Field verification required |

### 3.4 Capacity Analysis for Existing No-Build Condition

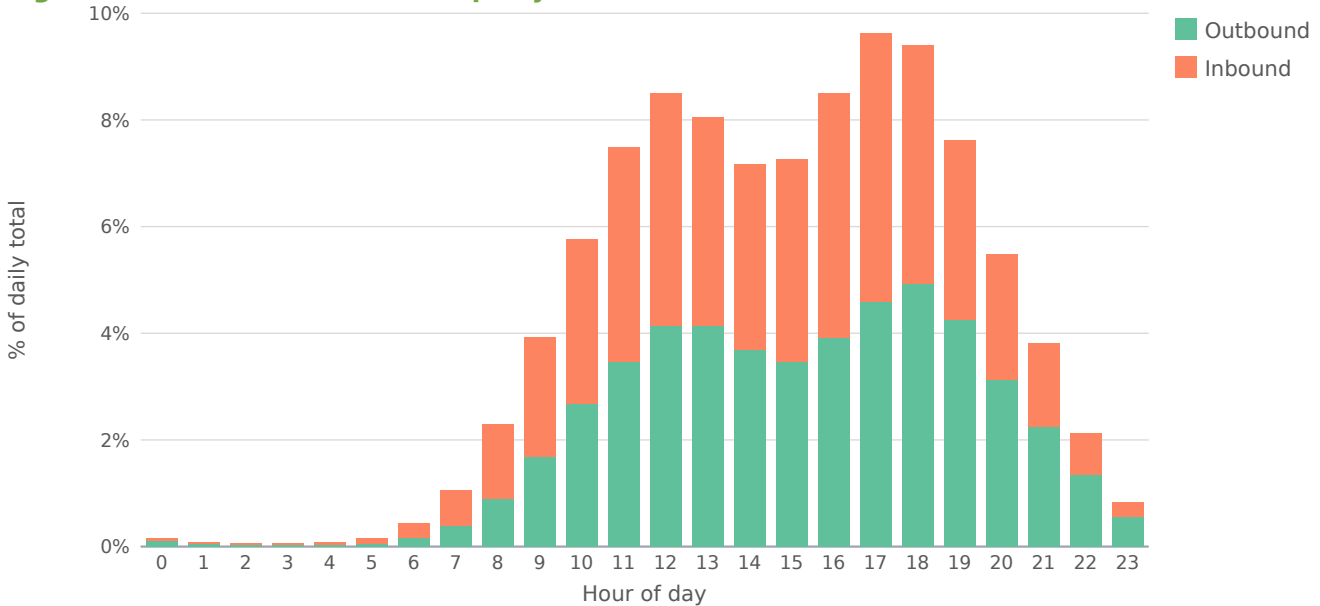
Existing No-Build conditions reflect current-year peak-hour volumes (no growth, no project trips). Existing measured volumes are grown to opening year 2027 at 1.5%/yr to establish the No-Build baseline for §3.5 comparison.

| Intersection                          | Existing LQS | Existing delay (s) | No-Build LQS | No-Build delay (s) | v/c  |
|---------------------------------------|--------------|--------------------|--------------|--------------------|------|
| Getzville Road & Sheridan Drive       | C            | 24.5               | C            | 24.9               | 0.70 |
| Near Sheridan Drive                   | C            | 24.5               | C            | 24.9               | 0.70 |
| Harlem Road & Sheridan Drive          | C            | 24.5               | C            | 24.9               | 0.70 |
| Near Harlem Road                      | B            | 16.6               | B            | 16.7               | 0.30 |
| Sheridan Drive & Youngmann Expressway | B            | 14.8               | B            | 14.8               | 0.13 |
| Millersport Highway & Sheridan Drive  | F            | 120.0              | F            | 120.0              | 1.23 |
| Millersport Highway & Sheridan Drive  | F            | 120.0              | F            | 120.0              | 1.23 |
| Sheridan Drive & Frankhauser Road     | F            | 120.0              | F            | 120.0              | 1.86 |
| Millersport Highway & Flint Road      | F            | 120.0              | F            | 120.0              | 1.36 |

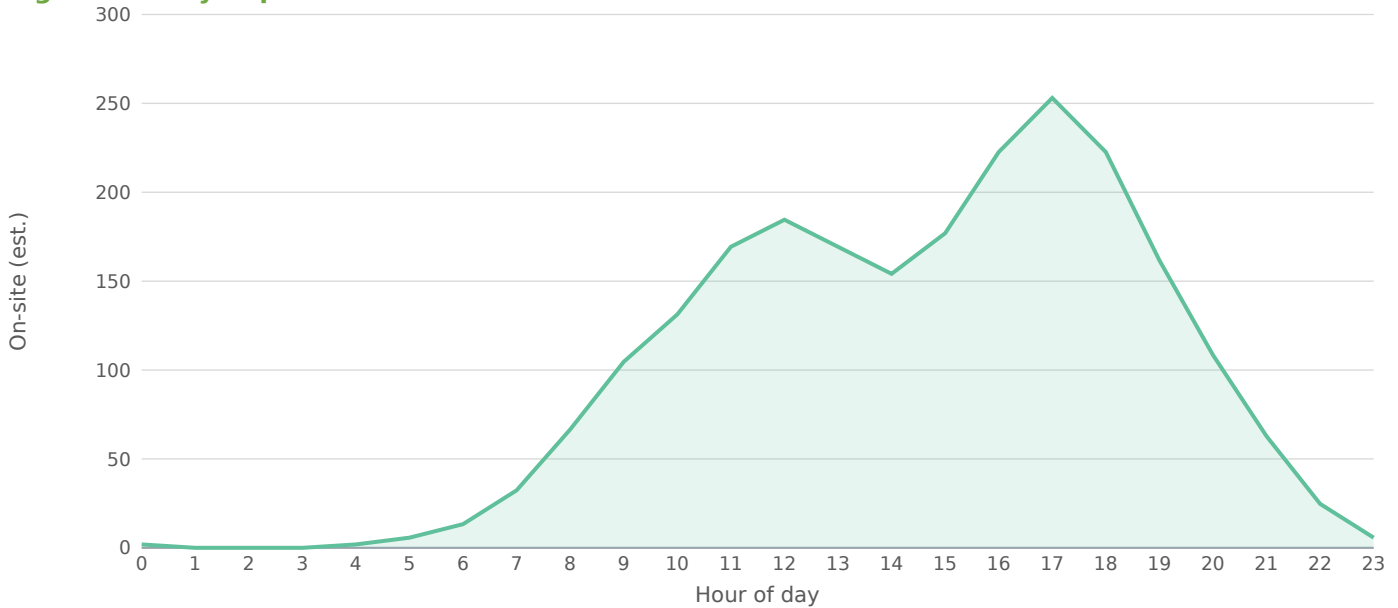
### Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

**Figure — Inbound / Outbound Trips by Start Time**



**Figure — Daily Trip Accumulation**



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

### 3.5 Capacity Analysis for Proposed Build Condition

Build conditions add the project's external peak-hour trips (680 PM peak, 326 inbound / 354 outbound) to the No-Build baseline at each affected intersection. Per HDM Chapter 5, intersection projects should also analyze each adjacent intersection plus any signal within ½ mile of the project site if expected left-turn volume changes materially affect capacity. The engine's network is determined by a 0.75-mile screening radius; manual scoping should confirm the ½-mile left-turn rule is satisfied.

| Intersection                         | ETC No-Build | ETC Build | ETC+20 NB | ETC+20 Bld | Δ delay (s) |
|--------------------------------------|--------------|-----------|-----------|------------|-------------|
| Getzville Road & Sheridan Drive      | C            | C         | D         | E          | 6.2         |
| Near Sheridan Drive                  | C            | C         | D         | E          | 5.1         |
| Harlem Road & Sheridan Drive         | C            | C         | D         | D          | 2.0         |
| Near Harlem Road                     | B            | B         | B         | B          | 0.6         |
| Sheridan Drive & Youngmann           | B            | B         | B         | B          | 0.3         |
| Expressway                           |              |           |           |            |             |
| Millersport Highway & Sheridan Drive | F            | F         | F         | F          | 0.0         |
| Millersport Highway & Sheridan Drive | F            | F         | F         | F          | 0.0         |
| Sheridan Drive & Frankhauser Road    | F            | F         | F         | F          | 0.0         |
| Millersport Highway & Flint Road     | F            | F         | F         | F          | 0.0         |

### 3.6 Capacity Improvement Measures

The following capacity improvements are recommended at affected intersections under Build conditions. Improvement category guidance per HDM Chapter 5: addition of turn lanes; installation or re-timing of traffic signals; signage and striping to prevent unwanted turning movements; conversion to roundabout; and TDM / TSM measures per HDM Chapter 24.

#### Getzville Road & Sheridan Drive [MINOR]

Minor: signal-timing optimization (shift 3-5s of green to the critical phase) is sufficient. No geometric change required.

#### Near Sheridan Drive [MINOR]

Minor: signal-timing optimization (shift 3-5s of green to the critical phase) is sufficient. No geometric change required.

#### Millersport Highway & Sheridan Drive [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

#### **Millersport Highway & Sheridan Drive [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

#### **Sheridan Drive & Frankhauser Road [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

#### **Millersport Highway & Flint Road [MAJOR]**

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

### **3.7 Trip Distribution — Gravity Model**

The trip distribution uses the gravity model (term =  $M / (d^{1.1} \cdot d_{site})$ ; mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

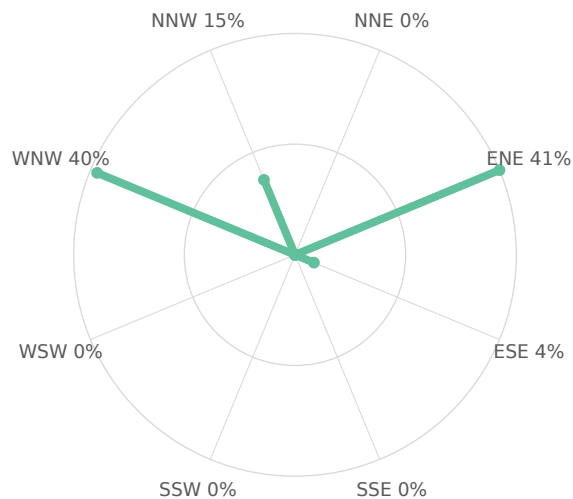
| Directional sector    |  | Share of project trips |  |  |  |
|-----------------------|--|------------------------|--|--|--|
| Northeast (NNE + ENE) |  | 41%                    |  |  |  |
| Southeast (ESE + SSE) |  | 4%                     |  |  |  |
| Southwest (SSW + WSW) |  | 0%                     |  |  |  |
| Northwest (WNW + NNW) |  | 55%                    |  |  |  |

| Study-area zone                       | Dir. | Distance (mi) | Mass (M) | Gravity term | Trip share |
|---------------------------------------|------|---------------|----------|--------------|------------|
| Sheridan Drive & Frankhauser Road     | ENE  | 0.68          | 3,301    | 0.2          | 20.98%     |
| Millersport Highway & Sheridan Drive  | WNW  | 0.44          | 2,182    | 0.2          | 15.17%     |
| Millersport Highway & Flint Road      | NNW  | 0.74          | 2,404    | 0.2          | 15.10%     |
| Millersport Highway & Sheridan Drive  | WNW  | 0.46          | 2,182    | 0.2          | 15.04%     |
| Getzville Road & Sheridan Drive       | WNW  | 0.13          | 1,242    | 0.1          | 9.67%      |
| Near Sheridan Drive                   | ENE  | 0.16          | 1,242    | 0.1          | 9.55%      |
| Harlem Road & Sheridan Drive          | ENE  | 0.31          | 1,242    | 0.1          | 9.02%      |
| Near Harlem Road                      | ESE  | 0.33          | 528      | 0.0          | 3.81%      |
| Sheridan Drive & Youngmann Expressway | ENE  | 0.42          | 236      | 0.0          | 1.65%      |

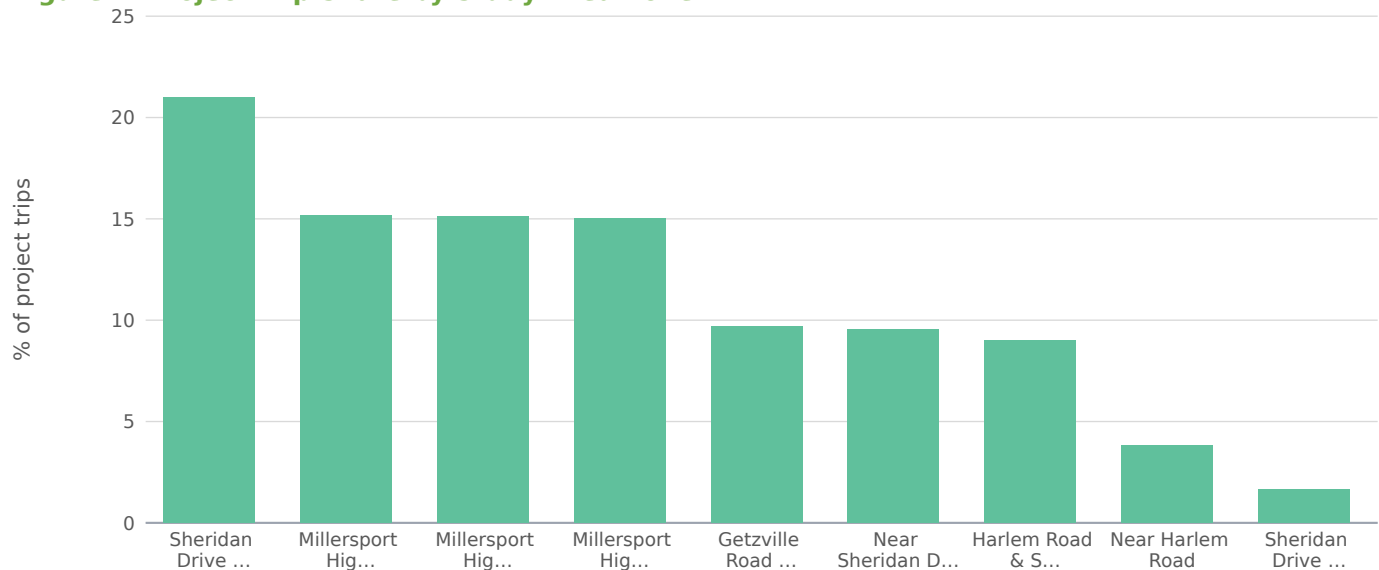
9 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

Figure — Directional Distribution of Project Trips



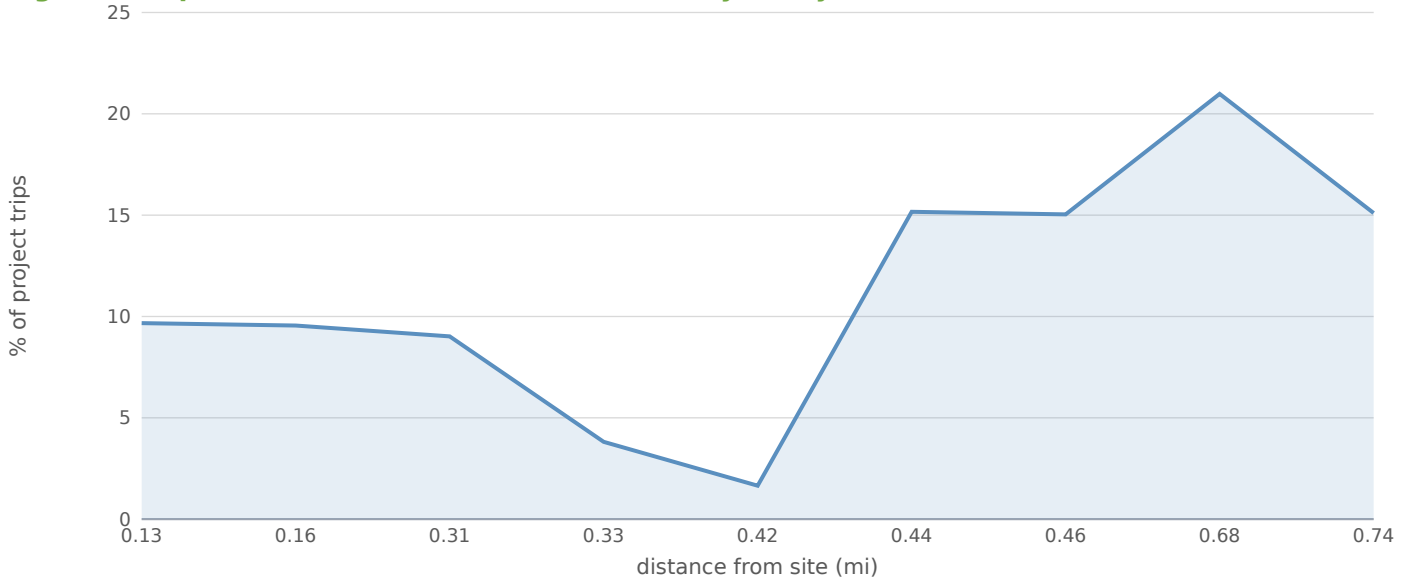
Screening-grade directional distribution of net new project trips by compass octant (spoke length  $\propto$  percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone





**Figure — Trip Share vs. Distance from Site (Gravity Decay)**



### 3.8 Project Trip Assignment

| Study intersection                    | Dir. | Distance (mi) | AM trips | PM trips | Share of project |
|---------------------------------------|------|---------------|----------|----------|------------------|
| Getzville Road & Sheridan Drive       | WNW  | 0.13          | 120      | 228      | 29.2%            |
| Near Sheridan Drive                   | ENE  | 0.16          | 104      | 197      | 25.3%            |
| Harlem Road & Sheridan Drive          | ENE  | 0.31          | 49       | 93       | 11.9%            |
| Near Harlem Road                      | ESE  | 0.33          | 46       | 87       | 11.2%            |
| Sheridan Drive & Youngmann Expressway | ENE  | 0.42          | 29       | 54       | 6.9%             |
| Millersport Highway & Sheridan Drive  | WNW  | 0.44          | 27       | 50       | 6.4%             |
| Millersport Highway & Sheridan Drive  | WNW  | 0.46          | 24       | 45       | 5.8%             |
| Sheridan Drive & Frankhauser Road     | ENE  | 0.68          | 8        | 15       | 1.9%             |
| Millersport Highway & Flint Road      | NNW  | 0.74          | 6        | 11       | 1.4%             |

Project trips assigned to each study intersection from the distribution shares above: the directional shares orient the loading toward the high-share sectors, distance-decayed to each approach.

### 4.0 CRASH ANALYSIS

A minimum three-year accident history review was performed for the period 2023 to 2026 per HDM Chapter 5 §5.3.4. County-level crash counts below are auto-ingested from the New York State Police "Motor Vehicle Crashes - Case Information: Three Year Window" Open Data dataset (<https://data.ny.gov/Transportation/Motor-Vehicle-Crashes-Case-Information-Three-Year-/e8ky-4vqe>), filtered by county\_name resolved from the project site coordinates via the NYSDOT RDM\_Roadway\_Current FeatureServer. The values cover the entire rolling 3-year reporting window for ERIE County, statewide-system-wide — they are NOT a per-segment or per-intersection rate.

| Severity descriptor (NY DMV) | Count (3-yr rolling) | Annualized |
|------------------------------|----------------------|------------|
|------------------------------|----------------------|------------|

| Severity descriptor (NY DMV)      | Count (3-yr rolling) | Annualized |
|-----------------------------------|----------------------|------------|
| Fatal Accident                    | 223                  | 74.3       |
| Injury Accident                   | 3,162                | 1054.0     |
| Property Damage Accident          | 55,706               | 18568.7    |
| Property Damage & Injury Accident | 19,889               | 6629.7     |
| Total (all severities)            | 78,980               | 26326.7    |

Within ERIE County over the rolling 3-year window, 78,980 crashes were reported to NY DMV (223 fatal, 23,051 injury-involved, 55,706 property damage only). The Fatal share is 0.094% of all reported crashes (annualized). For the Safety Screening per HDM Chapter 5 §5.3.4, the project-specific rate must be normalized against the controlling facility's exposure (segment AADT × segment length; intersection AADT × entry-flow) for comparison to NYSDOT Office of Modal Safety statewide averages and the 1.5× HAL trigger. County-totals above provide a county-baseline reference only — site-radius (1/4-mile typical) crash records, HAL / PIL / SDL flags, and CMF / CRF mitigation values must be obtained from the Region 5 — Buffalo / Western New York Regional Traffic Office via FOIL or SIMS. Required NYSDOT forms when the §4.0 section is expanded for formal submittal: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation), TE-204a (Accident Rate Summary), TE-213 (HAL / PIL / SDL Screening Output).

## APPENDIX A — EXISTING VOLUME REPORT

Per-approach existing peak-hour volumes (vph) for all affected intersections within study limits.  
Source: SimpleImpactStudies HCM 6 solver, calibrated against the controlling DOT 511 / Regional Traffic Office data feed.

| Intersection                          | Approach | Existing (vph) | No-Build (vph) | Build (vph) |
|---------------------------------------|----------|----------------|----------------|-------------|
| Getzville Road & Sheridan Drive       | NB       | 350            | 355            | 355         |
| Getzville Road & Sheridan Drive       | SB       | 297            | 301            | 327         |
| Getzville Road & Sheridan Drive       | EB       | 331            | 336            | 420         |
| Getzville Road & Sheridan Drive       | WB       | 265            | 269            | 387         |
| Near Sheridan Drive                   | NB       | 348            | 354            | 354         |
| Near Sheridan Drive                   | SB       | 338            | 343            | 343         |
| Near Sheridan Drive                   | EB       | 283            | 287            | 390         |
| Near Sheridan Drive                   | WB       | 273            | 278            | 372         |
| Harlem Road & Sheridan Drive          | NB       | 334            | 339            | 339         |
| Harlem Road & Sheridan Drive          | SB       | 352            | 357            | 357         |
| Harlem Road & Sheridan Drive          | EB       | 280            | 284            | 333         |
| Harlem Road & Sheridan Drive          | WB       | 276            | 280            | 325         |
| Near Harlem Road                      | NB       | 163            | 166            | 166         |
| Near Harlem Road                      | SB       | 130            | 132            | 132         |
| Near Harlem Road                      | EB       | 126            | 128            | 173         |
| Near Harlem Road                      | WB       | 109            | 111            | 153         |
| Sheridan Drive & Youngmann Expressway | NB       | 63             | 64             | 64          |
| Sheridan Drive & Youngmann Expressway | SB       | 65             | 66             | 66          |
| Sheridan Drive & Youngmann Expressway | EB       | 55             | 56             | 84          |
| Sheridan Drive & Youngmann Expressway | WB       | 53             | 54             | 80          |
| Millersport Highway & Sheridan Drive  | NB       | 706            | 717            | 717         |
| Millersport Highway & Sheridan Drive  | SB       | 498            | 505            | 511         |
| Millersport Highway & Sheridan Drive  | EB       | 501            | 509            | 527         |
| Millersport Highway & Sheridan Drive  | WB       | 477            | 484            | 510         |
| Millersport Highway & Sheridan Drive  | NB       | 672            | 682            | 682         |
| Millersport Highway & Sheridan Drive  | SB       | 621            | 630            | 634         |
| Millersport Highway & Sheridan Drive  | EB       | 485            | 492            | 509         |
| Millersport Highway & Sheridan Drive  | WB       | 405            | 411            | 434         |
| Sheridan Drive & Frankhauser Road     | NB       | 942            | 956            | 956         |
| Sheridan Drive & Frankhauser Road     | SB       | 863            | 876            | 876         |
| Sheridan Drive & Frankhauser Road     | EB       | 741            | 752            | 759         |
| Sheridan Drive & Frankhauser Road     | WB       | 755            | 766            | 773         |
| Millersport Highway & Flint Road      | NB       | 630            | 640            | 646         |
| Millersport Highway & Flint Road      | SB       | 678            | 688            | 690         |
| Millersport Highway & Flint Road      | EB       | 587            | 596            | 598         |

| Intersection                     | Approach | Existing (vph) | No-Build (vph) | Build (vph) |
|----------------------------------|----------|----------------|----------------|-------------|
| Millersport Highway & Flint Road | WB       | 509            | 517            | 519         |

## APPENDIX B — EXISTING CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Existing No-Build condition (current-year volumes; volumes grown to opening year at the §3.1 background-growth rate for No-Build comparison).

| Intersection                                    | Approach | Vol (vph) | v/c  | Delay (s) | LOS |
|-------------------------------------------------|----------|-----------|------|-----------|-----|
| Getzville Road & Sheridan Drive                 | NB       | 355       | 0.44 | 18.7      | B   |
| Getzville Road & Sheridan Drive                 | SB       | 301       | 0.37 | 17.7      | B   |
| Getzville Road & Sheridan Drive                 | EB       | 336       | 0.41 | 18.3      | B   |
| Getzville Road & Sheridan Drive                 | WB       | 269       | 0.33 | 17.1      | B   |
| Near Sheridan Drive                             | NB       | 354       | 0.44 | 18.7      | B   |
| Near Sheridan Drive                             | SB       | 343       | 0.42 | 18.4      | B   |
| Near Sheridan Drive                             | EB       | 287       | 0.35 | 17.4      | B   |
| Near Sheridan Drive                             | WB       | 278       | 0.34 | 17.2      | B   |
| Harlem Road & Sheridan Drive                    | NB       | 339       | 0.42 | 18.4      | B   |
| Harlem Road & Sheridan Drive                    | SB       | 357       | 0.44 | 18.7      | B   |
| Harlem Road & Sheridan Drive                    | EB       | 284       | 0.35 | 17.4      | B   |
| Harlem Road & Sheridan Drive                    | WB       | 280       | 0.35 | 17.3      | B   |
| Near Harlem Road                                | NB       | 166       | 0.20 | 15.6      | B   |
| Near Harlem Road                                | SB       | 132       | 0.16 | 15.1      | B   |
| Near Harlem Road                                | EB       | 128       | 0.16 | 15.1      | B   |
| Near Harlem Road                                | WB       | 111       | 0.14 | 14.9      | B   |
| Sheridan Drive & Youngmann                      | NB       | 64        | 0.08 | 14.3      | B   |
| Expressway Sheridan Drive & Youngmann           | SB       | 66        | 0.08 | 14.3      | B   |
| Expressway Sheridan Drive & Youngmann           | EB       | 56        | 0.07 | 14.2      | B   |
| Expressway Sheridan Drive & Youngmann           | WB       | 54        | 0.07 | 14.2      | B   |
| Expressway Millersport Highway & Sheridan Drive | NB       | 717       | 0.88 | 36.1      | D   |
| Millersport Highway & Sheridan Drive            | SB       | 505       | 0.62 | 22.5      | C   |
| Millersport Highway & Sheridan Drive            | EB       | 509       | 0.63 | 22.6      | C   |
| Millersport Highway & Sheridan Drive            | WB       | 484       | 0.60 | 21.9      | C   |
| Millersport Highway & Sheridan Drive            | NB       | 682       | 0.84 | 32.3      | C   |
| Millersport Highway & Sheridan Drive            | SB       | 630       | 0.78 | 28.2      | C   |
| Millersport Highway & Sheridan Drive            | EB       | 492       | 0.61 | 22.1      | C   |
| Millersport Highway & Sheridan Drive            | WB       | 411       | 0.51 | 19.9      | B   |
| Sheridan Drive & Frankhauser Road               | NB       | 956       | 1.18 | 118.5     | F   |
| Sheridan Drive & Frankhauser Road               | SB       | 876       | 1.08 | 80.6      | F   |
| Sheridan Drive & Frankhauser Road               | EB       | 752       | 0.93 | 41.7      | D   |
| Sheridan Drive & Frankhauser Road               | WB       | 766       | 0.95 | 44.6      | D   |
| Millersport Highway & Flint Road                | NB       | 640       | 0.79 | 28.8      | C   |
| Millersport Highway & Flint Road                | SB       | 688       | 0.85 | 32.8      | C   |
| Millersport Highway & Flint Road                | EB       | 596       | 0.74 | 26.2      | C   |
| Millersport Highway & Flint Road                | WB       | 517       | 0.64 | 22.9      | C   |

## APPENDIX C — PROPOSED CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Proposed Build condition (No-Build volumes plus project external trips at the assigned distribution).

| Intersection                          | Approach | Vol (vph) | v/c  | Delay (s) | LOS |
|---------------------------------------|----------|-----------|------|-----------|-----|
| Getzville Road & Sheridan Drive       | NB       | 355       | 0.44 | 18.7      | B   |
| Getzville Road & Sheridan Drive       | SB       | 327       | 0.40 | 18.1      | B   |
| Getzville Road & Sheridan Drive       | EB       | 420       | 0.52 | 20.1      | C   |
| Getzville Road & Sheridan Drive       | WB       | 387       | 0.48 | 19.4      | B   |
| Near Sheridan Drive                   | NB       | 354       | 0.44 | 18.7      | B   |
| Near Sheridan Drive                   | SB       | 343       | 0.42 | 18.4      | B   |
| Near Sheridan Drive                   | EB       | 390       | 0.48 | 19.4      | B   |
| Near Sheridan Drive                   | WB       | 372       | 0.46 | 19.0      | B   |
| Harlem Road & Sheridan Drive          | NB       | 339       | 0.42 | 18.4      | B   |
| Harlem Road & Sheridan Drive          | SB       | 357       | 0.44 | 18.7      | B   |
| Harlem Road & Sheridan Drive          | EB       | 333       | 0.41 | 18.2      | B   |
| Harlem Road & Sheridan Drive          | WB       | 325       | 0.40 | 18.1      | B   |
| Near Harlem Road                      | NB       | 166       | 0.20 | 15.6      | B   |
| Near Harlem Road                      | SB       | 132       | 0.16 | 15.1      | B   |
| Near Harlem Road                      | EB       | 173       | 0.21 | 15.7      | B   |
| Near Harlem Road                      | WB       | 153       | 0.19 | 15.4      | B   |
| Sheridan Drive & Youngmann Expressway | NB       | 64        | 0.08 | 14.3      | B   |
| Sheridan Drive & Youngmann Expressway | SB       | 66        | 0.08 | 14.3      | B   |
| Sheridan Drive & Youngmann Expressway | EB       | 84        | 0.10 | 14.5      | B   |
| Sheridan Drive & Youngmann Expressway | WB       | 80        | 0.10 | 14.5      | B   |
| Millersport Highway & Sheridan Drive  | NB       | 717       | 0.88 | 36.1      | D   |
| Millersport Highway & Sheridan Drive  | SB       | 511       | 0.63 | 22.7      | C   |
| Millersport Highway & Sheridan Drive  | EB       | 527       | 0.65 | 23.3      | C   |
| Millersport Highway & Sheridan Drive  | WB       | 510       | 0.63 | 22.7      | C   |
| Millersport Highway & Sheridan Drive  | NB       | 682       | 0.84 | 32.3      | C   |
| Millersport Highway & Sheridan Drive  | SB       | 634       | 0.78 | 28.4      | C   |
| Millersport Highway & Sheridan Drive  | EB       | 509       | 0.63 | 22.7      | C   |
| Millersport Highway & Sheridan Drive  | WB       | 434       | 0.54 | 20.5      | C   |
| Sheridan Drive & Frankhauser Road     | NB       | 956       | 1.18 | 118.5     | F   |
| Sheridan Drive & Frankhauser Road     | SB       | 876       | 1.08 | 80.6      | F   |
| Sheridan Drive & Frankhauser Road     | EB       | 759       | 0.94 | 43.2      | D   |
| Sheridan Drive & Frankhauser Road     | WB       | 773       | 0.95 | 46.2      | D   |
| Millersport Highway & Flint Road      | NB       | 646       | 0.80 | 29.2      | C   |
| Millersport Highway & Flint Road      | SB       | 690       | 0.85 | 33.0      | C   |
| Millersport Highway & Flint Road      | EB       | 598       | 0.74 | 26.3      | C   |

| Intersection                     | Approach | Vol (vph) | v/c  | Delay (s) | LOS |
|----------------------------------|----------|-----------|------|-----------|-----|
| Millersport Highway & Flint Road | WB       | 519       | 0.64 | 23.0      | C   |

## **APPENDIX D — CRASH ANALYSIS DIAGRAMS AND TABLES**

Crash analysis diagrams and tables are not produced by this screening analysis. When the §4.0 crash-analysis section is expanded for formal submittal, the following NYSDOT forms are required: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation Form), TE-204a (Accident Rate Summary), and TE-213 (HAL / PIL / SDL Screening Output). SIMS output and statewide-average rate references should be obtained from the controlling Regional Traffic Office per HDM Chapter 5 §5.3.4 and the NYSDOT Office of Modal Safety statewide rate tables (<https://www.dot.ny.gov/divisions/operating/osss/highway/accident-rates>).

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.



## FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

### Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

|                           |                                   |
|---------------------------|-----------------------------------|
| Land use                  | LU 820 — Shopping Center / Retail |
| Size                      | 85 1,000 sqft GLA                 |
| Daily trips (productions) | 6,800                             |
| PM peak-hour trips        | 680 (326 in / 354 out)            |

### Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals:  $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$ . Attractiveness  $A_j$  is each signal's through-volume; the friction factor  $F_j$  is the NCHRP-716 gamma function  $F = a \cdot t^b \cdot e^{(c \cdot t)}$  (home-based-work:  $a=28507$ ,  $b=-0.02$ ,  $c=-0.123$ ) on the travel time  $t$  to the signal.

| Signal (attraction zone)              | Dist mi | Time min | PM trips | Share |
|---------------------------------------|---------|----------|----------|-------|
| Getzville Road & Sheridan Drive       | 0.13    | 0.3      | 228      | 29.2% |
| Near Sheridan Drive                   | 0.16    | 0.4      | 197      | 25.3% |
| Harlem Road & Sheridan Drive          | 0.31    | 0.7      | 93       | 11.9% |
| Near Harlem Road                      | 0.33    | 0.8      | 87       | 11.2% |
| Sheridan Drive & Youngmann Expressway | 0.42    | 1.0      | 54       | 6.9%  |
| Millersport Highway & Sheridan Drive  | 0.44    | 1.1      | 50       | 6.4%  |
| Millersport Highway & Sheridan Drive  | 0.46    | 1.1      | 45       | 5.8%  |
| Sheridan Drive & Frankhauser Road     | 0.68    | 1.6      | 15       | 1.9%  |
| Millersport Highway & Flint Road      | 0.74    | 1.8      | 11       | 1.4%  |

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

### Step 3 — Mode Choice

Auto-mode share 94% applied via a binary logit  $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$  calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 6% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

### Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function  $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$  iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build  $v/c$  among study signals: 1.87. Per-signal  $v/c$ , delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

## APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

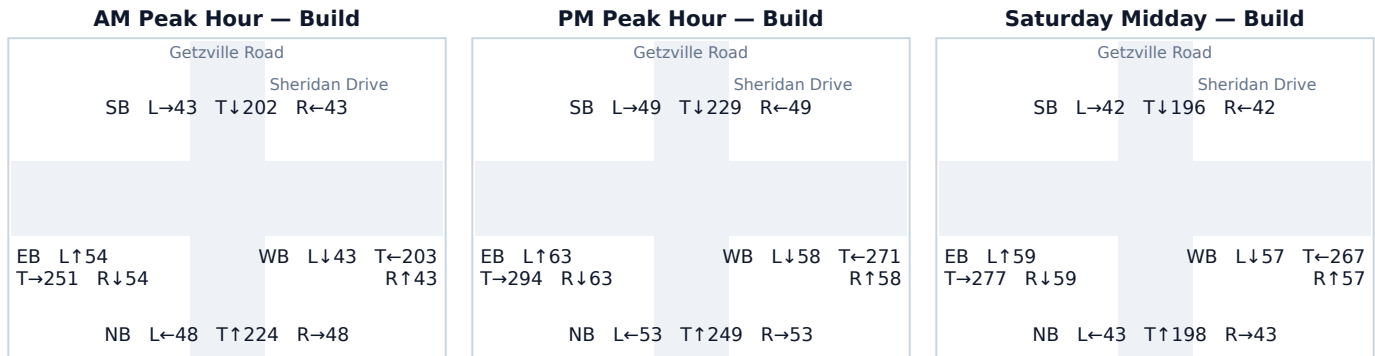
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

## A.1 Getzville Road & Sheridan Drive

|                          |                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | buffalo-1009                                                                                                               |
| Location                 | NW Buffalo-Cheektowaga-Niagara Falls · 0.13 mi from site                                                                   |
| Added PM peak trips      | 228                                                                                                                        |
| Existing / No-Build (PM) | LOS C · 24.9 s/veh · v/c 0.70                                                                                              |
| Build (PM)               | LOS C · 31.1 s/veh · v/c 0.83                                                                                              |
| Δ control delay          | +6.2 s/veh                                                                                                                 |
| Mitigation               | Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 355       | 0      | 355       | 0.44 → 0.44 | 18.7 → 18.7  | B → B      | 251    |
| SB    | 301       | 26     | 327       | 0.37 → 0.40 | 17.7 → 18.1  | B → B      | 227    |
| EB    | 336       | 84     | 420       | 0.41 → 0.52 | 18.3 → 20.1  | B → C      | 310    |
| WB    | 269       | 118    | 387       | 0.33 → 0.48 | 17.1 → 19.4  | B → B      | 280    |

### Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| WB Through | 90            | 39%                |
| EB Through | 84            | 37%                |
| WB Right   | 28            | 12%                |
| SB Left    | 26            | 11%                |

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.2 Near Sheridan Drive

Signal ID

Location

Added PM peak trips

Existing / No-Build (PM)

Build (PM)

Δ control delay

Mitigation

buffalo-1006

NW Buffalo-Cheektowaga-Niagara Falls · 0.16 mi from site

197

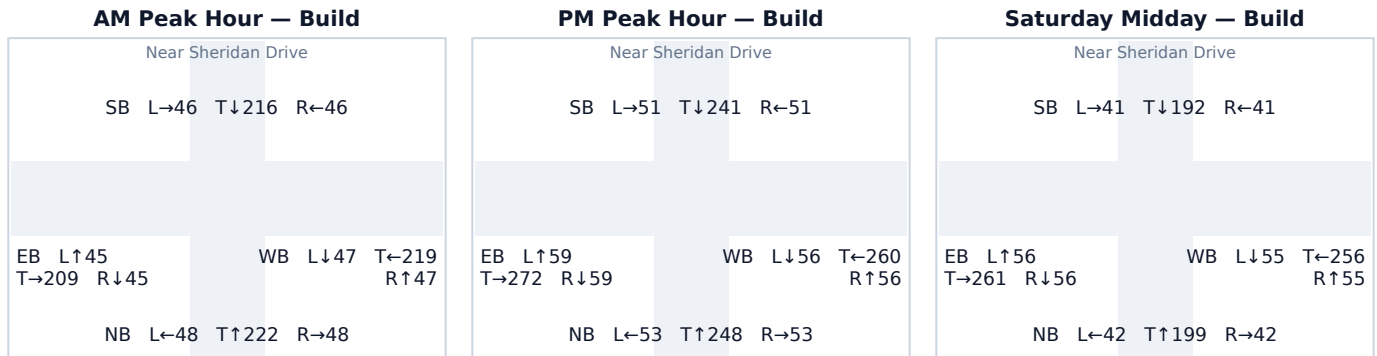
LOS C · 24.9 s/veh · v/c 0.70

LOS C · 30.0 s/veh · v/c 0.81

+5.1 s/veh

Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 354       | 0      | 354       | 0.44 → 0.44 | 18.7 → 18.7  | B → B      | 250    |
| SB    | 343       | 0      | 343       | 0.42 → 0.42 | 18.4 → 18.4  | B → B      | 240    |
| EB    | 287       | 102    | 390       | 0.35 → 0.48 | 17.4 → 19.4  | B → B      | 282    |
| WB    | 278       | 95     | 372       | 0.34 → 0.46 | 17.2 → 19.0  | B → B      | 266    |

### Affected movements (PM peak project trips)

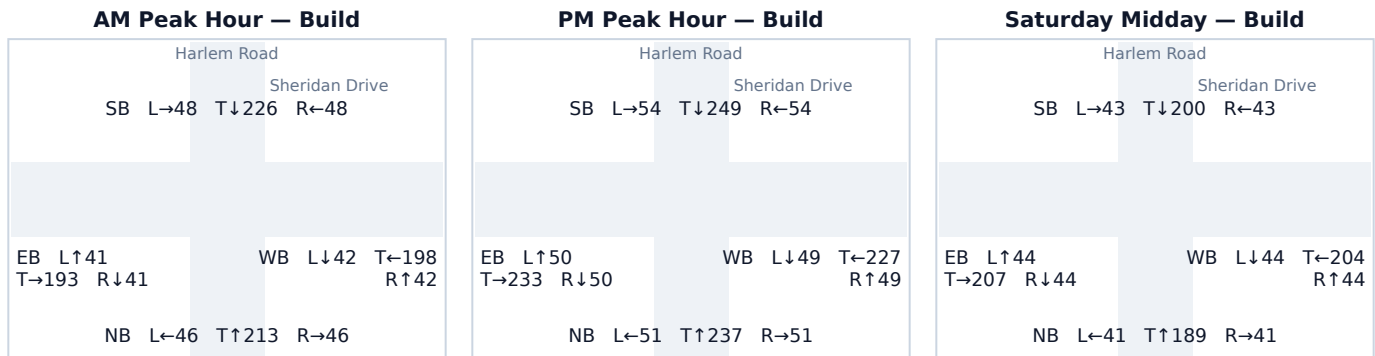
| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Through | 102           | 52%                |
| WB Through | 95            | 48%                |

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

### A.3 Harlem Road & Sheridan Drive

|                          |                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------|
| Signal ID                | buffalo-1007                                                                                |
| Location                 | NW Buffalo-Cheektowaga-Niagara Falls · 0.31 mi from site                                    |
| Added PM peak trips      | 93                                                                                          |
| Existing / No-Build (PM) | LOS C · 24.9 s/veh · v/c 0.70                                                               |
| Build (PM)               | LOS C · 26.9 s/veh · v/c 0.75                                                               |
| Δ control delay          | +2.0 s/veh                                                                                  |
| Mitigation               | No mitigation required — projected delay change is below the City's 5-second TIS threshold. |

#### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

#### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 339       | 0      | 339       | 0.42 → 0.42 | 18.4 → 18.4  | B → B      | 237    |
| SB    | 357       | 0      | 357       | 0.44 → 0.44 | 18.7 → 18.7  | B → B      | 253    |
| EB    | 284       | 48     | 333       | 0.35 → 0.41 | 17.4 → 18.2  | B → B      | 231    |
| WB    | 280       | 45     | 325       | 0.35 → 0.40 | 17.3 → 18.1  | B → B      | 225    |

#### Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Through | 48            | 52%                |
| WB Through | 45            | 48%                |

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.4 Near Harlem Road

Signal ID

Location

Added PM peak trips

Existing / No-Build (PM)

Build (PM)

Δ control delay

Mitigation

buffalo-781

NW Buffalo-Cheektowaga-Niagara Falls · 0.33 mi from site

87

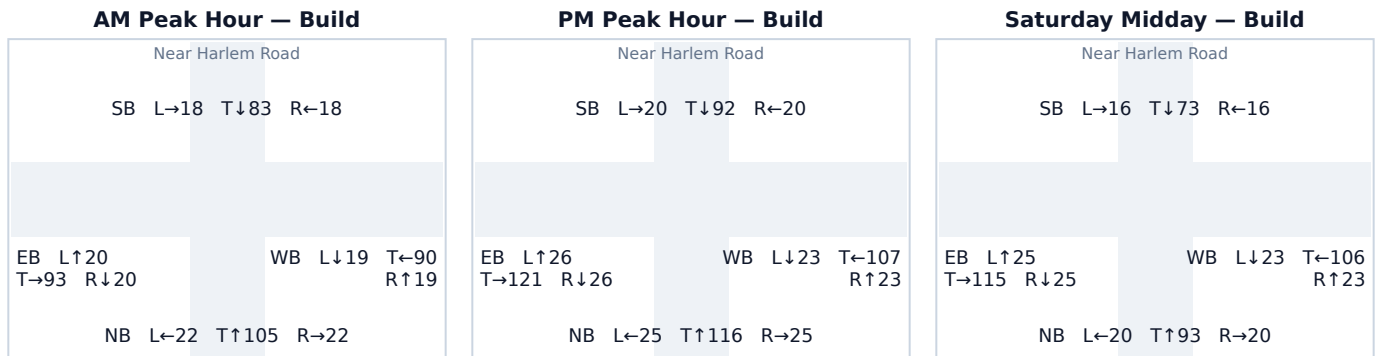
LOS B · 16.7 s/veh · v/c 0.30

LOS B · 17.3 s/veh · v/c 0.35

+0.6 s/veh

No mitigation required — projected delay change is below the City's 5-second TIS threshold.

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 166       | 0      | 166       | 0.20 → 0.20 | 15.6 → 15.6  | B → B      | 104    |
| SB    | 132       | 0      | 132       | 0.16 → 0.16 | 15.1 → 15.1  | B → B      | 81     |
| EB    | 128       | 45     | 173       | 0.16 → 0.21 | 15.1 → 15.7  | B → B      | 108    |
| WB    | 111       | 42     | 153       | 0.14 → 0.19 | 14.9 → 15.4  | B → B      | 95     |

### Affected movements (PM peak project trips)

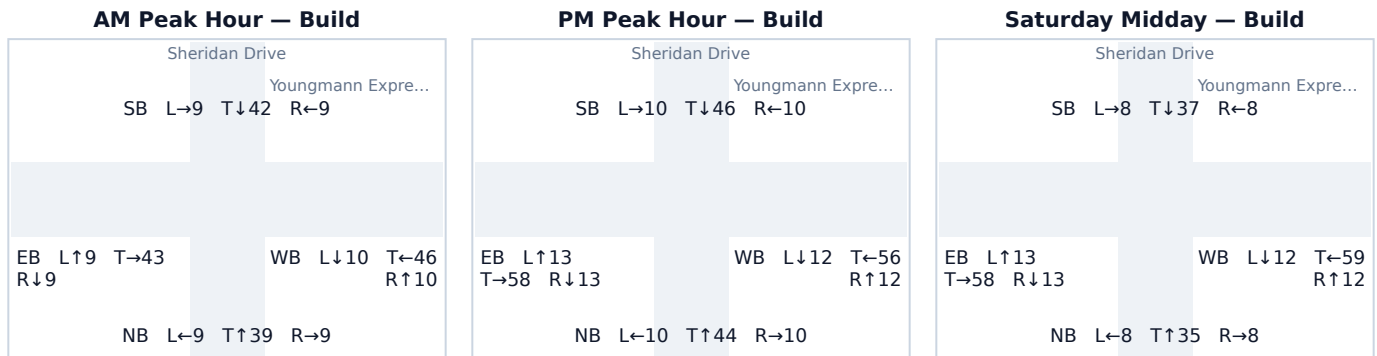
| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Through | 45            | 52%                |
| WB Through | 42            | 48%                |

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.5 Sheridan Drive & Youngmann Expressway

|                          |                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------|
| Signal ID                | buffalo-938                                                                                 |
| Location                 | NW Buffalo-Cheektowaga-Niagara Falls · 0.42 mi from site                                    |
| Added PM peak trips      | 54                                                                                          |
| Existing / No-Build (PM) | LOS B · 14.8 s/veh · v/c 0.13                                                               |
| Build (PM)               | LOS B · 15.1 s/veh · v/c 0.16                                                               |
| Δ control delay          | +0.3 s/veh                                                                                  |
| Mitigation               | No mitigation required — projected delay change is below the City's 5-second TIS threshold. |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 64        | 0      | 64        | 0.08 → 0.08 | 14.3 → 14.3  | B → B      | 38     |
| SB    | 66        | 0      | 66        | 0.08 → 0.08 | 14.3 → 14.3  | B → B      | 39     |
| EB    | 56        | 28     | 84        | 0.07 → 0.10 | 14.2 → 14.5  | B → B      | 50     |
| WB    | 54        | 26     | 80        | 0.07 → 0.10 | 14.2 → 14.5  | B → B      | 48     |

### Affected movements (PM peak project trips)

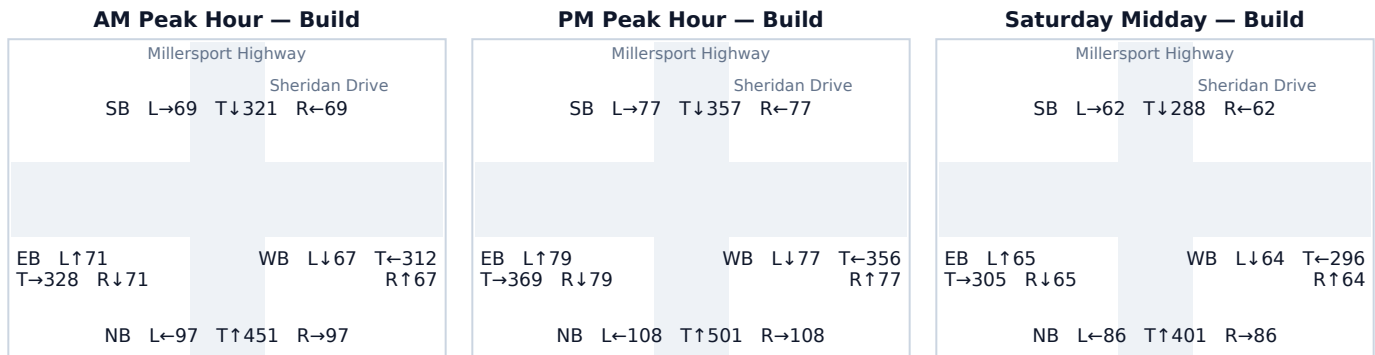
| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Through | 28            | 52%                |
| WB Through | 26            | 48%                |

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.6 Millersport Highway & Sheridan Drive

|                          |                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | buffalo-226                                                                                                                                                          |
| Location                 | NW Buffalo-Cheektowaga-Niagara Falls · 0.44 mi from site                                                                                                             |
| Added PM peak trips      | 50                                                                                                                                                                   |
| Existing / No-Build (PM) | LOS F · 120.0 s/veh · v/c 1.23                                                                                                                                       |
| Build (PM)               | LOS F · 120.0 s/veh · v/c 1.26                                                                                                                                       |
| Δ control delay          | +0.0 s/veh                                                                                                                                                           |
| Mitigation               | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 717       | 0      | 717       | 0.88 → 0.88 | 36.1 → 36.1  | D → D      | 675    |
| SB    | 505       | 5      | 511       | 0.62 → 0.63 | 22.5 → 22.7  | C → C      | 404    |
| EB    | 509       | 19     | 527       | 0.63 → 0.65 | 22.6 → 23.3  | C → C      | 423    |
| WB    | 484       | 26     | 510       | 0.60 → 0.63 | 21.9 → 22.7  | C → C      | 404    |

### Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| WB Through | 20            | 40%                |
| EB Through | 19            | 38%                |
| WB Right   | 6             | 12%                |
| SB Left    | 5             | 10%                |

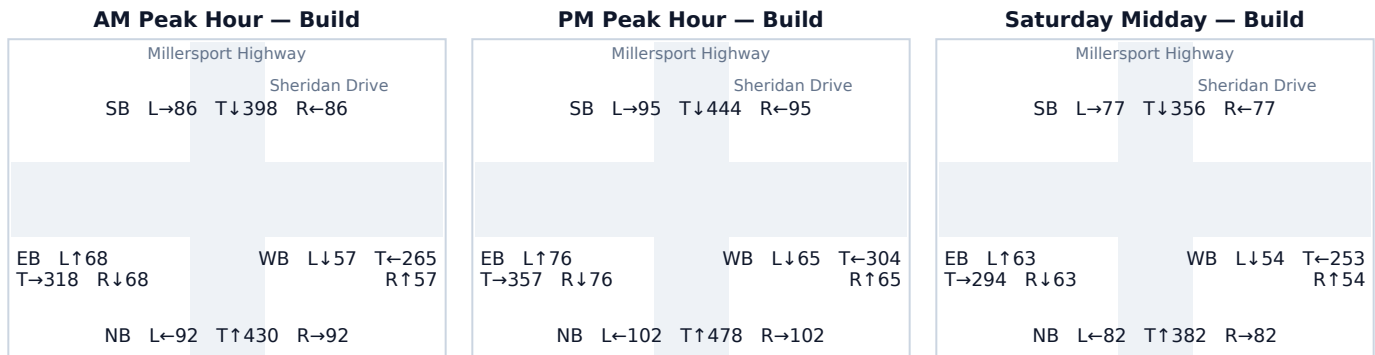
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.



## A.7 Millersport Highway & Sheridan Drive

|                          |                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | buffalo-325                                                                                                                                                          |
| Location                 | NW Buffalo-Cheektowaga-Niagara Falls · 0.46 mi from site                                                                                                             |
| Added PM peak trips      | 45                                                                                                                                                                   |
| Existing / No-Build (PM) | LOS F · 120.0 s/veh · v/c 1.23                                                                                                                                       |
| Build (PM)               | LOS F · 120.0 s/veh · v/c 1.26                                                                                                                                       |
| Δ control delay          | +0.0 s/veh                                                                                                                                                           |
| Mitigation               | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 682       | 0      | 682       | 0.84 → 0.84 | 32.3 → 32.3  | C → C      | 623    |
| SB    | 630       | 4      | 634       | 0.78 → 0.78 | 28.2 → 28.4  | C → C      | 555    |
| EB    | 492       | 18     | 509       | 0.61 → 0.63 | 22.1 → 22.7  | C → C      | 403    |
| WB    | 411       | 23     | 434       | 0.51 → 0.54 | 19.9 → 20.5  | B → C      | 324    |

### Affected movements (PM peak project trips)

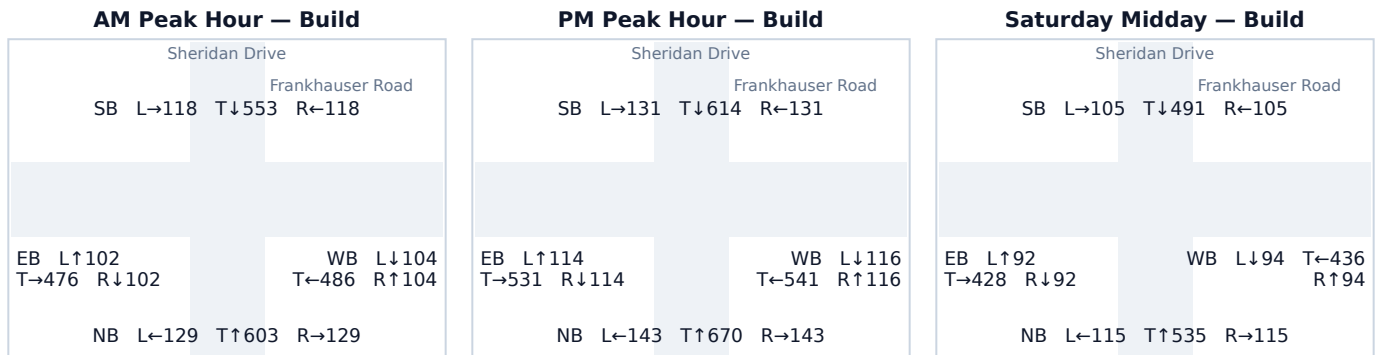
| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| WB Through | 19            | 42%                |
| EB Through | 18            | 40%                |
| WB Right   | 4             | 9%                 |
| SB Left    | 4             | 9%                 |

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.8 Sheridan Drive & Frankhauser Road

|                          |                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | buffalo-326                                                                                                                                                          |
| Location                 | NW Buffalo-Cheektowaga-Niagara Falls · 0.68 mi from site                                                                                                             |
| Added PM peak trips      | 15                                                                                                                                                                   |
| Existing / No-Build (PM) | LOS F · 120.0 s/veh · v/c 1.86                                                                                                                                       |
| Build (PM)               | LOS F · 120.0 s/veh · v/c 1.87                                                                                                                                       |
| Δ control delay          | +0.0 s/veh                                                                                                                                                           |
| Mitigation               | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld  | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|---------------|------------|--------|
| NB    | 956       | 0      | 956       | 1.18 → 1.18 | 118.5 → 118.5 | F → F      | 978    |
| SB    | 876       | 0      | 876       | 1.08 → 1.08 | 80.6 → 80.6   | F → F      | 896    |
| EB    | 752       | 8      | 759       | 0.93 → 0.94 | 41.7 → 43.2   | D → D      | 745    |
| WB    | 766       | 7      | 773       | 0.95 → 0.95 | 44.6 → 46.2   | D → D      | 769    |

### Affected movements (PM peak project trips)

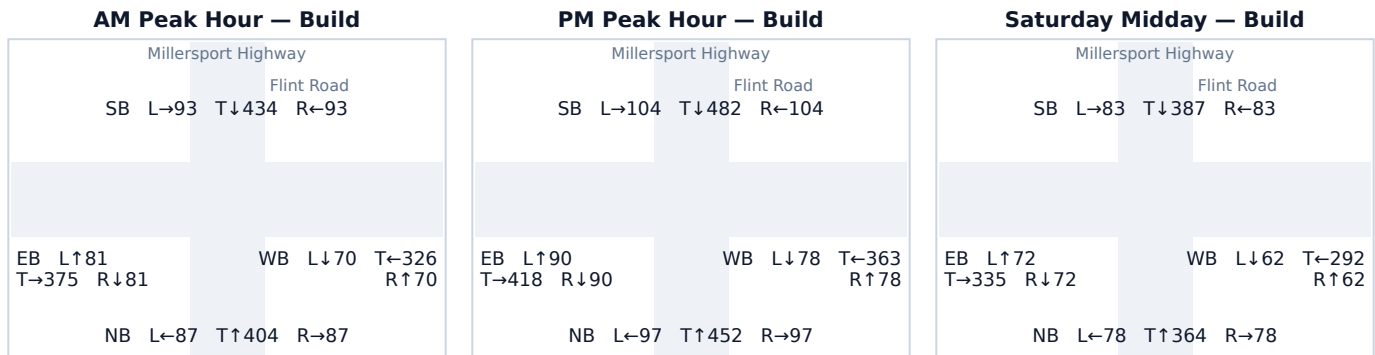
| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| EB Through | 8             | 53%                |
| WB Through | 7             | 47%                |

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

## A.9 Millersport Highway & Flint Road

|                          |                                                                                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal ID                | buffalo-173                                                                                                                                                          |
| Location                 | NW Buffalo-Cheektowaga-Niagara Falls · 0.74 mi from site                                                                                                             |
| Added PM peak trips      | 11                                                                                                                                                                   |
| Existing / No-Build (PM) | LOS F · 120.0 s/veh · v/c 1.36                                                                                                                                       |
| Build (PM)               | LOS F · 120.0 s/veh · v/c 1.36                                                                                                                                       |
| Δ control delay          | +0.0 s/veh                                                                                                                                                           |
| Mitigation               | Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. |

### Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

### Per-Approach Capacity (PM Peak)

| Appr. | Exist vph | +Trips | Build vph | v/c Ex→Bld  | Delay Ex→Bld | LOS Ex→Bld | Q95 ft |
|-------|-----------|--------|-----------|-------------|--------------|------------|--------|
| NB    | 640       | 6      | 646       | 0.79 → 0.80 | 28.8 → 29.2  | C → C      | 571    |
| SB    | 688       | 1      | 690       | 0.85 → 0.85 | 32.8 → 33.0  | C → C      | 634    |
| EB    | 596       | 2      | 598       | 0.74 → 0.74 | 26.2 → 26.3  | C → C      | 507    |
| WB    | 517       | 2      | 519       | 0.64 → 0.64 | 22.9 → 23.0  | C → C      | 413    |

### Affected movements (PM peak project trips)

| Movement   | Project trips | % of project trips |
|------------|---------------|--------------------|
| NB Left    | 2             | 18%                |
| EB Right   | 2             | 18%                |
| NB Right   | 2             | 18%                |
| NB Through | 2             | 18%                |
| WB Left    | 2             | 18%                |
| SB Through | 1             | 9%                 |

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-

foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.